

Final Resolution of the Real-Axis Fourier-Decay Problem for Rvachev's Up-Function

**Lacunary spectra, a corrected iterated-logarithm law, sharp Lambert– W peaks,
and natural versus shell-adaptive Cesàro gauges**

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Abstract

Let

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^h}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

be the Fourier transform of Rvachev's compactly supported up-function under the $e^{-2\pi i x \xi}$ convention. This article audits the original question, four independent first-wave analyses, the Aistleitner–Hofer–Larcher paper on lacunary trigonometric products, and the subsequent comparative audit. The apparently conflicting answers are reconciled by an exact dyadic factorization: a universal triangular denominator produces $\exp[-(\log x)^2/(2 \log 2)]$, while a Thue–Morse sine cocycle supplies a whole spectrum of power corrections. The typical, arithmetic-mean, root-mean-square, and uniform exponents are respectively

$$\begin{aligned} \kappa_0 &= 3.1514961294\dots, & \kappa_1 &= 2.7480700149\dots, \\ \kappa_2 &= 2.6514961294\dots, & \kappa_\infty &= 2.3590148791\dots \end{aligned}$$

We give a direct covariance and martingale calculation showing that the correct per-level variance of $\log |2 \sin \pi x|$ under doubling is $\pi^2/4$, not $\pi^2/2$. Thus the law-of-the-iterated-logarithm constants printed in the 2017 lacunary-products paper, and inherited by one first-wave report, must be divided by $\sqrt{2}$.

For the original arithmetic Cesàro question, the natural gauge

$$g_0(x) = A_1 x^{-\kappa_1} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

solves the requested two-sided boundedness and gives limit one at dyadic endpoints, but its unrestricted averages have a nonconstant dyadic-mantissa profile governed by a continuous singular eigenmeasure. This proves nonexistence of an unrestricted limit for every shell-shape-stabilizing gauge, in particular for the usual Hardy-field or logarithmico-exponential closed forms. The comparative audit stopped there and consequently overstated its scope. We close the literal existential problem: by progressively flattening the shell measures on finer finite partitions, while using the very steep baseline slope to preserve monotonicity, one obtains a positive C^∞ decreasing gauge with the all- t limit. A Carleman–Hoischen approximation of the reciprocal and its derivative upgrades it to a positive real-analytic, strictly decreasing gauge. It has the same rough asymptotic $g(x) = g_0(x)x^{o(1)}$, but its within-shell shape necessarily becomes more complex and does not stabilize.

Finally, for the actual largest value $M_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|$, we verify the sharp boundary-layer formula

$$\log M_k = -\frac{\log 2}{2}k(k+1) - (k+1)\log \pi + (k+1)\log \frac{\sqrt{3}}{2} - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\log 2} + \mathcal{O}((\log \log k)^2),$$

where $c = (\log 2)\sqrt{3/2}/\pi$. This includes a second log-normal suppression in $\log \log x$, and is distinct from every fixed-ray or mean law.

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1 The question and the final verdict

Set

$$\Phi(x) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^h}\right), \quad x \geq 0, \quad (1)$$

and let $t_0 > 0$ be fixed. The original question asks for a real-valued function g , preferably elementary, real-analytic, positive and eventually decreasing, for which[9, 10]

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g(x)} dx \longrightarrow 1 \quad (t \rightarrow \infty). \quad (C)$$

or at least for which the same expression stays between two positive constants:

$$0 < A < \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g(x)} dx < B < \infty \quad (t \geq t_1). \quad (B)$$

The adjective “decay rate” is ambiguous unless one specifies whether the quantity being measured is pointwise, typical, averaged in an L^q norm, or maximized over a dyadic annulus. The exact answer is therefore a dictionary, not a single equivalent.

Write

$$a := \log 2, \quad \mathcal{E}_\kappa(x) := x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2a}\right). \quad (2)$$

The four principal exponents are

$$\kappa_0 = \frac{3}{2} + \frac{\log \pi}{a} = 3.1514961294723188\dots, \quad (3)$$

$$\kappa_1 = \frac{1}{2} + \frac{\log(\pi/\rho_1)}{a} = 2.7480700148713340\dots, \quad (4)$$

$$\kappa_2 = \frac{\log(2\pi)}{a} = 2.6514961294723188\dots, \quad (5)$$

$$\kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a} = 2.3590148791117407\dots, \quad (6)$$

where ρ_1 is the Perron root of the weighted doubling operator in (35); numerically

$$\rho_1 = 0.661322602060565\dots \quad (7)$$

The decimal is a high-precision numerical value. Rigorous enclosures include $0.66130 < \rho_1 < 0.66135$ from Aistleitner–Hofer–Larcher and the independent exact-arithmetic certificate $0.66126807 < \rho_1 < 0.66134891$ in the fourth first-wave report[2, 14].

Theorem 1.1 (Final resolution). *The following statements hold.*

- (i) *There is no positive pointwise equivalent for $|\Phi(x)|$ on the full real ray, because every nonzero integer is a zero. Nevertheless*

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2 \log 2}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty.$$

- (ii) *On almost every fixed dyadic ray $x = 2^k y$, $1 < y < 2$, the exponent is κ_0 . The centered logarithm satisfies a CLT with variance $\pi^2/4$ per dyadic level and the LIL constants in (22). This corrects a factor $\sqrt{2}$ in the cited 2017 paper.*

- (iii) *The normalized L^q shell amplitudes form the pressure spectrum κ_q of (33). In particular, $q = 1$ gives the arithmetic-mean exponent κ_1 , $q = 2$ gives exactly $\kappa_2 = \log_2(2\pi)$, and $q = \infty$ gives κ_∞ .*
- (iv) *The natural analytic gauge $g_0 = A_1 \mathcal{E}_{\kappa_1}$, with $A_1 > 0$ defined in (46), satisfies (B); at $t = 2^k$ its Cesàro ratio tends to one. At $t = 2^k y$, the ratio tends to the nonconstant profile $\mathcal{P}(y)$ of (50). Consequently (C) fails for every shell-shape-stabilizing gauge, including all Hardy-field gauges of the correct rough size and their continuous log-periodic modulations.*
- (v) *In the literal unrestricted existential sense, (C) is affirmative. There exists a positive real-analytic strictly decreasing function g_{ad} on $(0, \infty)$ such that (C) holds and*

$$\log g_{\text{ad}}(x) = -\frac{(\log x)^2}{2 \log 2} - \kappa_1 \log x + o(\log x). \quad (8)$$

Its correction relative to g_0 is shell-adaptive and cannot have a stabilizing within-shell profile.

- (vi) *The natural gauge becomes fully convergent without adaptation if the arithmetic mean is replaced by the logarithmic mean dx/x , as in (53).*
- (vii) *The dyadic-annulus maximum obeys the Lambert- W asymptotic (79). Local peaks immediately following $m2^k$, with m fixed odd, obey the much smaller two-adic law (86).*

What “decays how fast” means. The universal factor is $\exp[-(\log x)^2/(2 \log 2)]$. Its power correction is $x^{-\kappa_0}$ at a typical dyadic phase, $x^{-\kappa_1}$ in absolute mean, $x^{-\kappa_2}$ in RMS, and $x^{-\kappa_\infty}$ for the sharp pointwise upper scale. The largest peak in an entire dyadic annulus has an additional Lambert- W , equivalently $(\log \log x)^2$, boundary-layer loss. For the original Cesàro question, a simple natural gauge gives boundedness but a nonconstant phase profile; an exact all-time analytic gauge exists only by allowing an increasingly nonstationary shell correction.

2 Audit of the supplied materials

The four first-wave documents are not mutually contradictory once their target functionals are separated. The substantive conflict is the LIL constant, and the substantive gap in the comparative audit is its extension of a class-restricted nonexistence theorem to the literal unrestricted question.

Source		Principal contribution	Final assessment
First-wave [11]	1	Exact dyadic reduction, almost-everywhere exponent, CLT/LIL, coarse and then refined annular behavior.	Its variance $\pi^2/4$ and LIL normalization are correct. It does not solve the literal arithmetic Cesàro question.
First-wave [12]	2	Strongest elementary treatment of the L^∞ geometry: calibrated Gelfond inequality, dense preimages of the maximizing cycle, Lambert- W annular maximum, and two-adic valleys.	Correct and sharply scoped. Its warning that the original L^1 question is separate is essential.

Source		Principal contribution	Final assessment
First-wave [13]	3	Arithmetic-mean exponent κ_1 , transfer operator, bounded Cesàro gauge, dyadic profile, and the q -spectrum viewpoint.	Correct on the mean theory, except that its LIL constant faithfully inherits the erroneous printed constant from the 2017 paper.
First-wave [14]	4	Most systematic L^q spectrum, exact L^2 identity, certificate for ρ_1 , singular eigenmeasure, unique lobe peaks, and exceptional valleys.	Correct where checked; the RMS identification $\kappa_2 = \log_2(2\pi)$ is exact. Its natural-gauge result is not a proof of nonexistence for arbitrary analytic gauges.
Aistleitner–Hofer–Larcher [2, 16]		Published lacunary-product LIL and the Fouvry–Mauduit constant ρ_1 .	The ρ_1 results are unaffected. Equation (4.5) omits the factor $1/2$ in cosine Parseval, doubling the variance. The printed LIL constants must be divided by $\sqrt{2}$.
Comparative audit [15]		Correct synthesis of the natural gauge, singular phase profile, class-restricted obstruction, logarithmic mean, and variance correction.	The theorem actually excludes shell-stabilizing gauges (and Hardy-field gauges), not every real-analytic decreasing gauge. The statement that the unrestricted limit is “unattainable in principle” is therefore too strong. Section 7 supplies a countervailing existence theorem.

Two numerical cautions are worth recording. First, digits such as $\rho_1 = 0.661322602060565\dots$ and $A_1 = 0.0912661\dots$ arise from spectral collocation and quadrature; the exact definitions, rather than the decimals, are used in the theorems. Second, the very slow approach to the asymptotic regime near two-adic valleys makes small-scale numerical fitting a poor guide to the final exponents.

3 Exact dyadic factorization and lobe structure

The infinite product converges locally uniformly, because $\text{sinc } z = 1 - z^2/6 + \mathcal{O}(z^4)$ at the origin. Hence Φ is even and entire. It also satisfies the exact refinement equation [1, 17]

$$\Phi(2x) = \text{sinc}(2\pi x)\Phi(x). \quad (9)$$

For $1 \leq y \leq 2$, put $t = y - 1$ and

$$P_n(t) := \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|, \quad (10)$$

$$H(y) := \prod_{r=1}^{\infty} \left| \text{sinc}\left(\frac{\pi y}{2^r}\right) \right|. \quad (11)$$

The tail H is continuous on $[1, 2]$, positive on $[1, 2)$, and $H(2) = 0$.

Proposition 3.1 (Exact shell factorization). *For every integer $k \geq 0$ and $1 \leq y \leq 2$, with $n = k + 1$,*

$$|\Phi(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-n} H(y) P_n(y - 1). \quad (12)$$

Proof. Split the factors in (1) at level k . In the first $k + 1$ factors, substitute $r = k - h$:

$$\prod_{h=0}^k \left| \operatorname{sinc} \left(\frac{\pi 2^k y}{2^h} \right) \right| = \frac{\prod_{r=0}^k |\sin(\pi 2^r y)|}{(\pi y)^{k+1} 2^{0+1+\dots+k}}.$$

Since $y = 1 + t$, the absolute numerator is $\prod_{r=0}^k |\sin(\pi 2^r t)| = P_{k+1}(t)$. The remaining factors are exactly $H(y)$. $\square$

The factorization isolates the deterministic triangular denominator from a multiplicative cocycle for the doubling map

$$Tt = 2t \pmod{1}, \quad P_n(t) = \prod_{j=0}^{n-1} s(T^j t), \quad s(t) = |\sin \pi t|. \quad (13)$$

It also gives the exact normalized identity

$$\frac{|\Phi(2^k y)|}{\mathcal{E}_\kappa(2^k y)} = C_\kappa(y) \exp(k(\kappa a - \log \pi - a/2)) P_{k+1}(y - 1), \quad (14)$$

where

$$C_\kappa(y) = \frac{H(y)}{\pi} y^{\kappa-1} \exp\left(\frac{(\log y)^2}{2a}\right). \quad (15)$$

All competing exponents arise by inserting a different exponential scale for P_n .

3.1 Zeros, signs, and one extremum per lobe

Euler's product for sine gives the canonical product

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2} \right)^{1+\nu_2(m)}. \quad (16)$$

Thus a nonzero integer m is a zero of multiplicity $1 + \nu_2(|m|)$. On every interval $(N, N + 1)$,

$$\frac{d^2}{dx^2} \log |\Phi(x)| = -2 \sum_{m=1}^{\infty} (1 + \nu_2(m)) \frac{m^2 + x^2}{(m^2 - x^2)^2} < 0. \quad (17)$$

The first derivative tends to $+\infty$ at the left endpoint and to $-\infty$ at the right endpoint. Hence every lobe has exactly one critical point, necessarily its strict maximum. The sign on $(N, N + 1)$ is $(-1)^{s_2(N)}$, the signed Thue–Morse value; the present article concerns its modulus.

4 Typical fixed rays and the corrected fluctuation law

Define the centered observable

$$X(t) := \log |2 \sin \pi t|, \quad S_n(t) := \sum_{j=0}^{n-1} X(T^j t). \quad (18)$$

Since $\int_0^1 \log |\sin \pi t| dt = -\log 2$, one has $\int_0^1 X = 0$. At $\kappa = \kappa_0$, (14) becomes the exact formula

$$|\Phi(2^k y)| = \mathcal{E}_{\kappa_0}(2^k y) D_0(y) e^{S_{k+1}(y-1)}, \quad D_0(y) = \frac{H(y)}{2\pi} y^{\kappa_0-1} e^{(\log y)^2/(2a)}. \quad (19)$$

Theorem 4.1 (Almost-everywhere dyadic-ray law). *For Lebesgue-almost every $y \in (1, 2)$,*

$$\log |\Phi(2^k y)| = -\frac{(\log(2^k y))^2}{2a} - \kappa_0 \log(2^k y) + o(k). \quad (20)$$

Moreover, with $n = k + 1$,

$$\frac{S_n}{\sqrt{n}} \Rightarrow N\left(0, \frac{\pi^2}{4}\right), \quad (21)$$

and almost surely

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} = \frac{\pi}{2}, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} = -\frac{\pi}{2}. \quad (22)$$

Equivalently, in the normalization $\sqrt{n \log \log n}$, the constants are $\pm\pi/\sqrt{2}$.

Proof. Birkhoff's theorem applied to the integrable potential $\log |\sin \pi t|$ gives $S_n/n \rightarrow 0$ almost everywhere, proving (20). The exact variance and martingale reduction are proved in the next two propositions; the standard stationary reverse-martingale CLT and LIL[5] then give (21)–(22). The coboundary remainder is negligible because X has exponentially decaying tails. $\square$

4.1 The exact covariance calculation

In $L^2(0, 1)$,

$$X(t) = -\sum_{m=1}^{\infty} \frac{\cos(2\pi m t)}{m}. \quad (23)$$

Cosine orthogonality gives, for $r \geq 0$,

$$c_r := \int_0^1 X(t) X(2^r t) dt = \frac{\pi^2}{12 \cdot 2^r}. \quad (24)$$

Indeed, only the frequency pairs $m = 2^r j$ survive, and every surviving cosine integral is $1/2$, not 1.

Proposition 4.2 (Exact finite variance). *For every $n \geq 1$,*

$$\text{Var}(S_n) = \frac{\pi^2}{4} n - \frac{\pi^2}{3} (1 - 2^{-n}). \quad (25)$$

In particular, the asymptotic variance per level is $\pi^2/4$.

Proof. Stationarity and (24) yield

$$\text{Var}(S_n) = n c_0 + 2 \sum_{r=1}^{n-1} (n-r) c_r.$$

Using $\sum_{r=1}^{n-1} (n-r) 2^{-r} = n - 2 + 2^{1-n}$ gives (25). $\square$

4.2 A one-line martingale decomposition

Let $\mathcal{P}$ be the Perron operator of doubling,

$$(\mathcal{P}h)(x) = \frac{1}{2} \left[h\left(\frac{x}{2}\right) + h\left(\frac{x+1}{2}\right) \right]. \quad (26)$$

The identity

$$\mathcal{P}X = \frac{1}{2}X \quad (27)$$

follows from $2 \sin(\pi x/2) 2 \cos(\pi x/2) = 2 \sin \pi x$. Set

$$d := 2X - X \circ T. \quad (28)$$

Then $\mathcal{P}d = 0$, and

$$S_n = \sum_{j=0}^{n-1} d \circ T^j + X \circ T^n - X. \quad (29)$$

The first term is a stationary reverse-martingale sum. Furthermore

$$\int_0^1 d^2 = 4c_0 + c_0 - 4c_1 = \frac{\pi^2}{4}. \quad (30)$$

Since $\text{Leb}(|X| > u) = \mathcal{O}(e^{-u})$, Borel–Cantelli implies $X \circ T^n = o(\sqrt{n \log \log n})$ almost surely. This proves that the martingale variance and LIL constants pass unchanged to S_n .

4.3 The factor- $\sqrt{2}$ error in the lacunary-products paper

The cited Aistleitner–Hofer–Larcher source[2, 16] evaluates the same variance functional as

$$\sum_{j \geq 1} \left(\frac{1}{j^2} + 2 \sum_{r \geq 1} \frac{1}{2^r j^2} \right) = \frac{\pi^2}{2}.$$

That displayed sum is twice the correct Parseval expression: each pairing of cosines contributes $1/2$. Therefore the source's equation (4.5) should be $\sigma_{f_1}^2 = \pi^2/4$. In the source's normalizations, its intermediate LIL constant $\pi/\sqrt{2}$ becomes $\pi/2$, its theorem constant π becomes $\pi/\sqrt{2}$, and the corresponding product constant $\pi/\sqrt{\log 2}$ becomes $\pi/\sqrt{2 \log 2}$. The Fouvry–Mauduit constant and the paper's L^1 -based discrepancy exponent are unaffected.

5 The full L^q decay spectrum

For $q > 0$, define the weighted transfer operator

$$(\mathcal{L}_q h)(x) = \frac{1}{2} \sum_{Ty=x} s(y)^q h(y), \quad s(y) = |\sin \pi y|. \quad (31)$$

Thus

$$\int_0^1 P_n(t)^q h(t) dt = \int_0^1 \mathcal{L}_q^n h(x) dx. \quad (32)$$

Let ρ_q be the spectral radius of $\mathcal{L}_q$, and put

$$\Lambda_q := \rho_q^{1/q}, \quad \kappa_q := \frac{\log \pi + a/2 - \log \Lambda_q}{a}. \quad (33)$$

Standard Ruelle–Perron–Frobenius theory on a suitable Hölder or bounded-variation space[4] gives positive eigenvectors and exponential convergence for the endpoint-power weights occurring here. It follows from (14) that the normalized L^q shell norm of $|\Phi|/\mathcal{E}_{\kappa_q}$ tends to a finite positive constant.

The endpoint cases can be computed almost entirely by hand.

5.1 Geometric mean and $q \downarrow 0$

Lebesgue measure is invariant under T , so

$$\int_0^1 \log P_n(t) dt = n \int_0^1 \log |\sin \pi t| dt = -na. \quad (34)$$

Hence $\Lambda_0 = 1/2$ and κ_0 is (3). This is the same exponent as the almost-everywhere fixed-ray law.

5.2 Arithmetic mean and the Fouvry–Mauduit constant

For $q = 1$, (31) becomes

$$(\mathcal{L}_1 h)(x) = \frac{1}{2} \left[\sin\left(\frac{\pi x}{2}\right) h\left(\frac{x}{2}\right) + \cos\left(\frac{\pi x}{2}\right) h\left(\frac{x+1}{2}\right) \right]. \quad (35)$$

Fouvry and Mauduit proved[3, 2]

$$I_1(n) := \int_0^1 P_n(t) dt = K \rho_1^n (1 + o(1)), \quad K > 0, \quad (36)$$

which fixes the arithmetic-mean exponent κ_1 .

5.3 Root mean square: an exact identity

Proposition 5.1 (Exact L^2 norm). *For every $n \geq 1$,*

$$\int_0^1 P_n(t)^2 dt = 2^{-n}. \quad (37)$$

Proof. Use $\sin^2 u = (1 - \cos 2u)/2$:

$$P_n(t)^2 = 2^{-n} \prod_{j=0}^{n-1} (1 - \cos(2\pi 2^j t)).$$

After expansion and product-to-sum, every nonconstant frequency is a signed sum of distinct powers of two. Uniqueness of binary representation prevents such a sum from being zero. Thus only the constant term survives integration. $\square$

Consequently $\rho_2 = 1/2$, $\Lambda_2 = 2^{-1/2}$, and $\kappa_2 = \log_2(2\pi)$. This explains the exponent proposed in the pre-wave draft: it is exactly the RMS exponent, not the pointwise or absolute mean exponent.

5.4 Uniform norm and the maximizing two-cycle

Put $u_j = \sin^2(\pi 2^j t)$, so $u_{j+1} = 4u_j(1 - u_j)$, and define $V(u) = 1 + 2u/3$. The calibrated subaction underlying Gel'fond's bound[6] is the algebraic identity

$$3V(u) - 4uV(4u(1 - u)) = \frac{(2u + 1)(4u - 3)^2}{3} \geq 0 \quad (38)$$

implies

$$|\sin(\pi 2^j t)| \leq \frac{\sqrt{3}}{2} \left(\frac{V(u_j)}{V(u_{j+1})} \right)^{1/2}. \quad (39)$$

Telescoping gives

$$P_n(t) \leq \sqrt{\frac{5}{3}} \left(\frac{\sqrt{3}}{2} \right)^n. \quad (40)$$

At $t = 1/3$ or $2/3$, every factor equals $\sqrt{3}/2$. Hence

$$\Lambda_\infty = \frac{\sqrt{3}}{2}, \quad \kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a}. \quad (41)$$

Combining (40) with (14) yields

$$|\Phi(x)| \leq C\mathcal{E}_{\kappa_\infty}(x) \quad (x \geq 1), \quad (42)$$

and equality in exponential scale on the fixed dyadic rays corresponding to the period-two orbit, for example $x = 2^k(4/3)$.

6 The natural arithmetic-mean gauge

Let

$$g_1(x) := \mathcal{E}_{\kappa_1}(x). \quad (43)$$

By (14) and the defining relation for κ_1 , there is a continuous bounded mantissa weight W_1 , positive on $[0, 1]$, such that

$$\frac{|\Phi(2^k(1 + z))|}{g_1(2^k(1 + z))} = \rho_1^{-(k+1)} W_1(z) P_{k+1}(z). \quad (44)$$

RPF convergence implies that the finite measures on $[0, 1]$

$$d\eta_k(z) := \rho_1^{-(k+1)} W_1(z) P_{k+1}(z) dz \quad (45)$$

converge weakly to a finite nonzero measure η . Define

$$A_1 := \eta([0, 1]) = \lim_{k \rightarrow \infty} \int_0^1 \frac{|\Phi(2^k(1 + z))|}{g_1(2^k(1 + z))} dz. \quad (46)$$

A spectral computation gives $A_1 \approx 0.09126612$, but only the exact limit definition is needed. Set

$$g_0(x) := A_1 g_1(x). \quad (47)$$

The measure $\mu := \eta/A_1$ is a probability measure. It is atomless, fully supported, and singular with respect to Lebesgue measure. The atomlessness and singularity are important enough to record explicitly.

Proposition 6.1 (The limiting shell measure is continuous singular). *The probability measure μ has no atoms, has full support on $[0, 1]$, and satisfies $\mu \perp \text{Leb}$.*

Proof. The RPF eigenmeasure ν for $\mathcal{L}_1^*$ obeys, at a point z , $\rho_1 \nu\{z\} = \frac{1}{2} s(z) \nu\{Tz\}$. Since $\rho_1 > 1/2$, a nonzero atom would force its forward masses to grow by a factor at least $2\rho_1 > 1$, contradicting finiteness. Topological mixing of doubling and positivity of the weight in the interior give full support. Multiplication by the continuous positive factor W_1 preserves these properties.

For singularity, suppose the eigenmeasure had a nonzero absolutely continuous part $r \, dz$. The dual eigen-equation gives almost everywhere

$$s(z)r(Tz) = \rho_1 r(z). \quad (48)$$

Thus

$$r(T^n z) = \frac{\rho_1^n}{P_n(z)} r(z). \quad (49)$$

For Lebesgue-almost every z , Birkhoff gives $P_n(z) = \exp[-n(a + o(1))]$, so on $\{r > 0\}$ the right side grows like $(2\rho_1)^n$. On the other hand, invariance of Lebesgue measure and Markov's inequality give

$$\text{Leb}\{r \circ T^n > e^{\varepsilon n}\} = \text{Leb}\{r > e^{\varepsilon n}\} \leq \|r\|_1 e^{-\varepsilon n}.$$

Borel–Cantelli contradicts geometric growth when $0 < \varepsilon < \log(2\rho_1)$. The Lebesgue decomposition is preserved by the dual operator, so the entire absolutely continuous part must vanish. $\square$

6.1 Dyadic endpoints, the phase profile, and boundedness

For $0 \leq z \leq 1$, let

$$\mathcal{F}(z) := \mu([0, z]), \quad \mathcal{P}(y) := \frac{1 + \mathcal{F}(y-1)}{y}, \quad 1 \leq y \leq 2. \quad (50)$$

The distribution function $\mathcal{F}$ is continuous and singular, so $\mathcal{P}$ is continuous but not identically one.

Theorem 6.2 (Natural-gauge Cesàro profile). *For every fixed $y \in [1, 2]$,*

$$\frac{1}{2^k y - t_0} \int_{t_0}^{2^k y} \frac{|\Phi(x)|}{g_0(x)} \, dx \longrightarrow \mathcal{P}(y). \quad (51)$$

In particular, the limit is one at $y = 1, 2$, and the ratio is eventually bounded above and below by positive constants for unrestricted t .

Proof. On a full shell, the change of variables $x = 2^k(1+z)$ and (44) show that the shell contribution divided by 2^k tends to one. The sum of all completed shells below 2^k is therefore $2^k + o(2^k)$. The partial current shell up to $z = y - 1$ is $2^k(\mathcal{F}(z) + o(1))$, by weak convergence and atomlessness. Division by $2^k y$ proves (51). Continuity and positivity of $\mathcal{P}$ on the compact interval $[1, 2]$ give the two-sided bound. $\square$

The audit's numerical profile range, approximately $[0.9249, 1.1133]$, is consistent with this theorem but is not used in the proof.

6.2 What the singular profile really rules out

Theorem 6.3 (No exact limit for stabilizing shell shapes). *Suppose $g > 0$ has the same rough shell scale as g_1 , and there are constants $c_k > 0$ and a continuous $u : [0, 1] \rightarrow (0, \infty)$ such that*

$$\frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \longrightarrow u(z) \quad \text{uniformly for } z \in [0, 1]. \quad (52)$$

Then (C) cannot hold for g .

Proof. If (C) held, subtraction of its asymptotics at $2^k(1+z)$ and 2^k would force the normalized mass of every partial shell $[0, z]$ to converge to z . By (52) and RPF convergence, those partial masses instead converge, after a scalar normalization, to $q d\mu$, where q is continuous and strictly positive on $[0, 1]$. Uniform linearity would therefore imply $q d\mu = dz$. This is impossible because μ is singular and atomless. $\square$

The class includes every Hardy-field gauge of the correct scale for which the within-shell logarithmic derivative has a finite asymptotic slope, and also such gauges multiplied by a continuous positive function of $\{\log_2 x\}$. Degenerate infinite slopes concentrate the shell mass at an edge and fail the required linear partial-shell law even more directly. This covers the usual logarithmico-exponential closed forms. It does *not* cover every real-analytic decreasing function: analyticity permits shell-dependent corrections whose complexity increases with k .

6.3 The logarithmic average repairs the natural gauge

Theorem 6.4 (Unrestricted logarithmic mean). *There is a constant $A_1^{\log} > 0$ such that*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|\Phi(x)|}{A_1^{\log} g_1(x)} \frac{dx}{x} \longrightarrow 1 \quad (t \rightarrow \infty) \quad (53)$$

without any restriction on the dyadic mantissa of t .

Proof. The logarithmic mass of the k -th shell converges by RPF to a positive constant. Summing K completed shells gives a quantity asymptotic to that constant times K , while the terminal partial shell is uniformly bounded. Since $\log t = Ka + \mathcal{O}(1)$, ordinary Cesàro summation proves the result. The terminal shell has relative weight $\mathcal{O}(1/\log t)$, so its singular within-shell distribution disappears. $\square$

7 A literal all-time analytic gauge: shell-adaptive flattening

This section closes the only remaining logical gap. The natural gauge cannot have an all-time arithmetic Cesàro limit because its normalized shell measures tend to a singular measure. But the original problem asks for *some* analytic decreasing gauge, not necessarily a gauge whose shell shape stabilizes. The deterministic baseline is so steep that a slowly increasing amount of within-shell reweighting can be inserted without making the total reciprocal gauge decrease.

Let

$$D_k(z) := \frac{|\Phi(2^k(1+z))|}{g_0(2^k(1+z))}, \quad d\mu_k(z) := D_k(z) dz. \quad (54)$$

By construction and Theorem 6.2,

$$\mu_k \implies \mu, \quad \mu_k([0, 1]) \longrightarrow 1, \quad (55)$$

where μ is atomless and fully supported.

7.1 A flattening lemma for weakly convergent shell measures

Lemma 7.1 (Smooth asymptotic flattening). *Let μ_k be finite positive Borel measures on $[0, 1]$ converging weakly to an atomless, fully supported probability measure μ , with $\mu_k([0, 1]) \rightarrow 1$. Then there are functions $r_k \in C^\infty([0, 1])$, $r_k > 0$, such that*

(a) $r_k \equiv 1$ in neighborhoods of 0 and 1;

(b)

$$\sup_{0 \leq z \leq 1} \left| \int_0^z r_k d\mu_k - z \right| \rightarrow 0; \quad (56)$$

(c)

$$\|\log r_k\|_\infty = o(k), \quad \|(\log r_k)'\|_\infty = o(k). \quad (57)$$

Proof. We give the diagonal construction because it is the mechanism that defeats the stabilizing-class obstruction.

For each integer $j \geq 1$, choose a finite partition $0 = x_{j,0} < x_{j,1} < \dots < x_{j,N_j} = 1$ with mesh at most $\varepsilon_j \downarrow 0$, and with all boundary points non-atoms of μ (automatic here). Full support gives $\mu(I_{j,i}) > 0$ for every partition interval $I_{j,i} = [x_{j,i-1}, x_{j,i}]$. Define limiting cell multipliers

$$c_{j,i} := \frac{|I_{j,i}|}{\mu(I_{j,i})}.$$

Choose a finite union U_j of very small intervals around all partition boundaries, including 0, 1, so that both $\text{Leb}(U_j)$ and $\mu(U_j)$ are as small as needed below and $\mu(\partial U_j) = 0$.

For sufficiently large k , all cell masses $\mu_k(I_{j,i})$ are close to $\mu(I_{j,i})$, and $\mu_k(U_j)$ is close to $\mu(U_j)$. Put

$$c_{k,j,i} := \frac{|I_{j,i}|}{\mu_k(I_{j,i})}.$$

On the portion of each cell outside U_j , set $\ell_{k,j} = \log c_{k,j,i}$. Join the finitely many constants smoothly inside U_j , and arrange $\ell_{k,j} = 0$ near 0 and 1. For fixed j , the values $c_{k,j,i}$ stay in a compact subinterval of $(0, \infty)$, so the interpolation can be chosen with uniform bounds

$$\|\ell_{k,j}\|_\infty \leq B_j, \quad \|\ell'_{k,j}\|_\infty \leq C_j \quad (k \geq K_j^{(0)}).$$

Let $r_{k,j} = e^{\ell_{k,j}}$. Away from U_j , the weighted mass of a whole cell is exactly its Lebesgue length. The discrepancy is confined to U_j , where the multiplier is uniformly bounded. Taking U_j small enough therefore gives, for all sufficiently large k ,

$$\sup_z \left| \int_0^z r_{k,j} d\mu_k - z \right| \leq 3\varepsilon_j.$$

The mesh bounds the possible error in the one partially traversed cell.

Choose increasing integers K_j so large that all preceding conclusions hold for $k \geq K_j$, and additionally

$$\frac{B_j}{K_j} \leq \varepsilon_j, \quad \frac{C_j}{K_j} \leq \varepsilon_j.$$

For $K_j \leq k < K_{j+1}$, set $r_k = r_{k,j}$. Then $j = j(k) \rightarrow \infty$, so the cumulative discrepancy tends to zero and both ratios in (57) tend to zero. The endpoint condition was built into every interpolation. $\square$

7.2 A positive smooth decreasing exact gauge

Define a global multiplier R on all sufficiently large x by

$$R(2^k(1+z)) := r_k(z), \quad 0 \leq z \leq 1. \quad (58)$$

Because every r_k equals one in neighborhoods of both endpoints, these pieces agree as the constant function one in neighborhoods of every dyadic boundary. Thus R is globally C^∞ .

Let

$$h_0(x) := \frac{1}{g_0(x)}, \quad h(x) := h_0(x)R(x), \quad g_\infty(x) := \frac{1}{h(x)}. \quad (59)$$

In shell coordinates,

$$\frac{\partial}{\partial z} \log h_0(2^k(1+z)) = \frac{k + \kappa_1 + a^{-1} \log(1+z)}{1+z}. \quad (60)$$

The right side is at least $k/3$ for all large k , uniformly in $z \in [0, 1]$. By (57), $\|(\log r_k)'\|_\infty = o(k)$; hence h is eventually strictly increasing and g_∞ is eventually strictly decreasing. Also

$$\log g_\infty(x) = \log g_0(x) + o(\log x). \quad (61)$$

Proposition 7.2 (Exact arithmetic mean for the smooth adaptive gauge). *The gauge g_∞ satisfies*

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g_\infty(x)} dx \longrightarrow 1 \quad (t \rightarrow \infty). \quad (62)$$

Proof. Let

$$C_k(z) := \int_0^z D_k(u) r_k(u) du.$$

Lemma 7.1 gives $\sup_z |C_k(z) - z| \rightarrow 0$. If $t = 2^K(1+z)$, then, up to a fixed initial contribution,

$$\int_{t_0}^t \frac{|\Phi(x)|}{g_\infty(x)} dx = \sum_{k < K} 2^k C_k(1) + 2^K C_K(z).$$

Replacing $C_k(1)$ by 1 and $C_K(z)$ by z creates an $o(2^K)$ error: this is the elementary Toeplitz fact that $\sum_{k < K} 2^k \varepsilon_k = o(2^K)$ whenever $\varepsilon_k \rightarrow 0$. The main term is $\sum_{k < K} 2^k + 2^K z = 2^K(1+z) + \mathcal{O}(1) = t + \mathcal{O}(1)$. This proves (62). $\square$

7.3 Real-analytic upgrade by Carleman–Hoischen approximation

We use the following classical approximation theorem in its C^1 form: for every real-valued $F \in C^1(\mathbb{R})$ and every positive continuous ϵ , there is an entire function G , real on $\mathbb{R}$, such that

$$|G(x) - F(x)| < \epsilon(x), \quad |G'(x) - F'(x)| < \epsilon(x) \quad (x \in \mathbb{R}). \quad (63)$$

This is Hoischen's derivative version of Carleman approximation; a modern proof is due to Gauthier and Kienzle[8].

Extend h from (59) to a positive strictly increasing C^1 function on all of $\mathbb{R}$. Choose a continuous $\delta(x) > 0$ with $\delta(x) \rightarrow 0$ as $x \rightarrow +\infty$, and apply (63) with

$$\epsilon(x) < \frac{1}{2} \min\{h(x), h'(x), \delta(x)h(x)\}. \quad (64)$$

Then the real entire approximant $\tilde{h}$ satisfies $\tilde{h} > 0$, $\tilde{h}' > 0$ on $\mathbb{R}$, and $\tilde{h}/h \rightarrow 1$ as $x \rightarrow +\infty$. Define

$$g_{\text{ad}}(x) := \frac{1}{\tilde{h}(x)}, \quad x > 0. \quad (65)$$

It is positive, real-analytic, and strictly decreasing.

Theorem 7.3 (Literal analytic solution of the original limit problem). *The gauge g_{ad} in (65) satisfies (C) and (8).*

Proof. The rough asymptotic follows from $\tilde{h}/h \rightarrow 1$, (61), and the definition of g_0 . For the integral, given $\varepsilon > 0$, choose X so that $|\tilde{h}/h - 1| < \varepsilon$ for $x \geq X$. Then

$$\left| \int_X^t |\Phi|(\tilde{h} - h) dx \right| \leq \varepsilon \int_X^t |\Phi|h dx = \varepsilon(t + o(t))$$

by Proposition 7.2. The initial interval contributes $\mathcal{O}(1)$. Letting $\varepsilon \downarrow 0$ proves the same limit for $\tilde{h} = 1/g_{\text{ad}}$. $\square$

Remark 7.4 (Why this does not contradict the natural obstruction). The multiplier r_k corrects only finitely many cells at level k , but the number of cells tends to infinity. Its logarithm and logarithmic slope are $o(k)$, so the total gauge remains decreasing and retains the κ_1 rough exponent. Nevertheless its shell shape never converges. Theorem 6.3 therefore does not apply. This also explains why no fixed power of x , fixed power of $\log x$, or continuous log-periodic multiplier can produce the exact arithmetic limit.

Remark 7.5 (On the optional higher-derivative request). The natural gauge g_0 has alternating derivative signs eventually for each fixed order. The adaptive analytic construction above guarantees positivity and strict decrease, which are the mandatory regularity conditions in the original wording. It does not claim one common tail on which every higher derivative is monotone. Preserving a prescribed finite number of higher derivative signs is possible by using the corresponding higher-order Carleman–Höischen theorem and slowing the diagonal construction further.

8 The sharp pointwise and annular envelopes

The pointwise upper scale $\mathcal{E}_{\kappa_\infty}(x)$ is sharp along fixed maximizing rays, but the largest value in the moving annulus $[2^k, 2^{k+1}]$ is a subtler boundary-layer problem. A fixed phase such as $y = 4/3$ incurs an exponential loss relative to the left endpoint scale. The true maximizer approaches the zero at 2^k , spends a slowly growing number of doubling steps in the small-angle regime, and then shadows the maximizing cycle $\{1/3, 2/3\}$.

Put

$$M_k := \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|, \quad n = k + 1, \quad \beta := \log \frac{\sqrt{3}}{2}, \quad (66)$$

and

$$c := \frac{a}{\pi} \sqrt{\frac{3}{2}}. \quad (67)$$

8.1 Boundary-layer reduction

Writing $x = 2^k(1 + \theta)$ in (12) gives

$$|\Phi(2^k(1 + \theta))| = 2^{-k(k+1)/2} \pi^{-n} H(1 + \theta) (1 + \theta)^{-n} P_n(\theta). \quad (68)$$

The symmetry $P_n(\theta) = P_n(1 - \theta)$ reduces the supremum, up to a constant independent of n , to $0 \leq \theta \leq 1/2$. Indeed, if $1/2 \leq \theta \leq 1$, then $H(2 - \theta)$ is bounded below while $H(1 + \theta)$ is bounded above, and $(2 - \theta)/(1 + \theta) \leq 1$. Consequently the value at θ is at most a fixed multiple of the value at $1 - \theta$.

For $0 < \theta \leq 1/2$, write

$$\theta = 2^{-r}s, \quad r \in \mathbb{Z}_{\geq 0}, \quad s \in I := [1/4, 1/2]. \quad (69)$$

If $r < n$, then

$$P_n(2^{-r}s) = \left[\prod_{h=1}^r \sin\left(\frac{\pi s}{2^h}\right) \right] P_{n-r}(s). \quad (70)$$

Uniformly for $s \in I$, the initial block satisfies

$$\log \prod_{h=1}^r \sin\left(\frac{\pi s}{2^h}\right) = -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} \right) + \mathcal{O}(1). \quad (71)$$

The Gelfond inequality gives the upper tail $P_m(s) \leq Ce^{m\beta}$. Conversely, the q -fold preimages of $1/3$ and $2/3$ form an $O(2^{-q})$ -net. For example, if $s_q = (j + 1/3)/2^q$, then $T^q s_q = 1/3$, and for $0 \leq \ell < q$

$$\text{dist}(T^\ell s_q, \mathbb{Z}) \geq \frac{1}{3 \cdot 2^{q-\ell}}.$$

Since $|\sin \pi u| \geq 2 \text{dist}(u, \mathbb{Z})$, the first q factors are bounded below by

$$\prod_{\ell=0}^{q-1} |\sin(\pi 2^\ell s_q)| \geq \left(\frac{2}{3}\right)^q 2^{-q(q+1)/2}.$$

All later factors lie exactly on the maximizing two-cycle. Hence a nearby preimage can be chosen with

$$P_m(s_q) \geq \exp(m\beta - Cq^2), \quad m \geq q. \quad (72)$$

At the relevant transition indices $r = \log_2 n + O(\log \log n)$, the finite objective is $O(\log n)$ -Lipschitz in s . Taking $q = \lceil 2 \log_2 \log n \rceil$ therefore makes the net-discretization loss $o(1)$, while the initial-orbit loss is $O(q^2)$. This is the final $O((\log \log n)^2)$ error term.

The finite objective is

$$\mathcal{F}_n(r, s) = -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} - \beta \right) - n \log(1 + s2^{-r}). \quad (73)$$

The preceding estimates give

$$\log \sup_{0 \leq \theta \leq 1} H(1 + \theta)(1 + \theta)^{-n} P_n(\theta) = n\beta + \sup_{r,s} \mathcal{F}_n(r, s) + \mathcal{O}((\log \log n)^2). \quad (74)$$

8.2 Lambert optimization

Set

$$C_0 := \log \pi - \frac{a}{2} - \beta = \log \left(\pi \sqrt{\frac{2}{3}} \right). \quad (75)$$

Replacing the final logarithm in (73) by its linear term changes the supremum by $o(1)$. Put $A = C_0 + \log s$ and $z = ar - A$. As s ranges over $[1/4, 1/2]$, A ranges over an interval of length exactly a ; as r runs through the integers, the corresponding z -intervals tile the real line. Up to a bounded term, the objective becomes

$$-\frac{z^2}{2a} - ne^{-C_0} e^{-z}. \quad (76)$$

Its critical equation is

$$ze^z = ane^{-C_0} = cn. \quad (77)$$

Thus $z = W(cn)$, with the principal Lambert function[7], and at the maximum

$$\sup_{r,s} \mathcal{F}_n(r,s) = -\frac{W(cn)^2 + 2W(cn)}{2a} + \mathcal{O}(1). \quad (78)$$

Theorem 8.1 (Sharp dyadic-annulus maximum). *As $k \rightarrow \infty$, with $n = k + 1$,*

$$\log M_k = -\frac{a}{2}k(k+1) - n \log \pi + n\beta - \frac{W(cn)^2 + 2W(cn)}{2a} + \mathcal{O}((\log \log k)^2). \quad (79)$$

Expanding $W(cn) = \log n - \log \log n + \log c + \mathcal{O}(\log \log n / \log n)$ gives

$$\begin{aligned} \log M_k = & -\frac{a}{2}k^2 - \left(\frac{a}{2} + \log \pi - \beta\right)k - \frac{(\log k)^2}{2a} + \frac{\log k \log \log k}{a} \\ & - \frac{1 + \log c}{a} \log k + \mathcal{O}((\log \log k)^2). \end{aligned} \quad (80)$$

In physical scale $R = 2^k$, the extra correction begins with

$$-\frac{(\log \log R)^2}{2 \log 2} + \frac{\log \log R \log \log \log R}{\log 2} + \mathcal{O}(\log \log R). \quad (81)$$

No fixed factor $(\log R)^{-p}$ can represent this boundary layer. The near-maximizing phase satisfies

$$\theta_k \asymp \frac{W(c(k+1))}{k+1}, \quad x_k = 2^k \left(1 + \Theta\left(\frac{\log k}{k}\right)\right). \quad (82)$$

9 Exceptional local peaks and two-adic valleys

Let E_N denote the unique maximum of $|\Phi|$ on $(N, N+1)$. Fix an odd positive integer m , put $N = m2^k$, and write $x = N + t$, $0 \leq t \leq 1$. The first $k+1$ factors yield uniformly

$$|\Phi(m2^k + t)| = (m2^k)^{-(k+1)} |H(m)| t^{k+1} \Phi(t) (1 + o(1)). \quad (83)$$

Near $t = 1$,

$$\Phi(t) = H(1)(1-t) + \mathcal{O}((1-t)^2). \quad (84)$$

The elementary boundary maximum $\max_{0 \leq t \leq 1} t^p(1-t) \sim 1/(ep)$ now gives the exact valley law.

Theorem 9.1 (Peaks after a fixed odd multiple of a power of two). *Let $c_{m2^k} = m2^k + t_k$ be the unique critical point in $(m2^k, m2^k + 1)$. Then*

$$(k+1)(1-t_k) \longrightarrow 1, \quad (85)$$

and

$$E_{m2^k} \sim \frac{H(1)|H(m)|}{e(k+1)} (m2^k)^{-(k+1)}. \quad (86)$$

In particular,

$$E_{2^k} \sim \frac{H(1)^2}{e(k+1)} 2^{-k(k+1)}. \quad (87)$$

The leading quadratic coefficient in $\log E_{2^k}$ is $-\log 2$, twice the magnitude of the annular maximum's coefficient $-\frac{1}{2} \log 2$. Thus even the sequence of local maxima has no single two-sided log-normal-power envelope.

10 Implications, limitations, and formalization targets

10.1 What is now completely settled

The following conclusions are rigorous, with their external analytic inputs explicitly identified.

- The exact factorization, the $q = 0$, $q = 2$, and $q = \infty$ rows, the variance correction, the Gelfond inequality, the Lambert optimization, and the two-adic valley asymptotic are elementary consequences of the product plus standard real analysis.
- The $q = 1$ exponent uses the Fouvry–Mauduit asymptotic (36). The shell profile uses the standard RPF spectral theorem for $\mathcal{L}_1$.
- The analytic adaptive gauge additionally uses the C^1 Carleman–Hoischen approximation theorem. Its shell-flattening construction is otherwise elementary once weak convergence $\mu_k \Rightarrow \mu$ is known.
- The literal original existence question (C) and the weaker question (B) are both affirmative. What fails is not existence, but existence inside the asymptotically stationary closed-form class.

10.2 What should not be claimed

There is no single pointwise equivalent, no universal exponent independent of the chosen norm, and no elementary justification for replacing the lacunary numerator by a typical mean when studying maxima. The high-precision decimals for ρ_1 , A_1 , and the phase profile are numerical, although the operators and limiting constants are exact. The analytic adaptive gauge is an existence construction, not a short elementary formula. Finally, the higher-derivative monotonicity requested only optionally in the original question is not part of Theorem 7.3.

10.3 Lean-friendly decomposition

A formal development can be divided into the following blocks.

1. Locally uniform convergence, refinement, zeros, and the exact shell factorization.
2. Strict log-concavity and uniqueness of every interzero maximum.
3. Exact L^2 identity and the calibrated polynomial inequality (38).
4. Fourier covariance calculation proving $\sigma^2 = \pi^2/4$.
5. Rational preimages and the finite Lambert objective.
6. Boundary maximum lemma and two-adic valleys.
7. Imported abstractions for Birkhoff, martingale CLT/LIL, Fouvry–Mauduit, and RPF.
8. The shell-flattening diagonal lemma, which is measure-theoretic and independent of the special product after $\mu_k \Rightarrow \mu$ is supplied. The final analytic upgrade may remain as an external complex-approximation theorem.

11 Conclusion

The Fourier transform of Rvachev's up-function does have an exact dominant decay scale:

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right).$$

Everything beyond that universal factor records how the lacunary sine cocycle is sampled. A typical orbit, an L^1 average, an L^2 average, and an extremal orbit yield four distinct powers. The largest peak in a moving dyadic annulus is more delicate still, acquiring a second log-normal layer in $\log \log x$. The correct fluctuation variance is $\pi^2/4$, resolving the only numerical disagreement among the first-wave analyses and exposing a Parseval slip in the cited 2017 paper.

For the exact question originally asked, the distinction between a *natural* gauge and an arbitrary analytic gauge is decisive. The natural transfer-operator gauge is explicit up to one spectral constant, solves the bounded problem, and displays a singular dyadic phase profile. That profile rules out every asymptotically stationary closed form. It does not rule out analyticity itself. By allowing the gauge to adapt on successively finer shell partitions—slowly enough that the universal log-square slope still dominates—one obtains a positive analytic decreasing gauge with the full all-time Cesàro limit. This is the final literal resolution: the desired analytic function exists, has the arithmetic-mean exponent κ_1 , but cannot be represented by a fixed elementary correction to the log-normal baseline.

A Constants and conversions

For convenience,

$$\begin{aligned} a &= \log 2 = 0.6931471805599453094\dots, \\ \beta &= \log(\sqrt{3}/2) = -0.1438410362258904637\dots, \\ c &= \frac{a}{\pi} \sqrt{\frac{3}{2}} = 0.2702223197333653409\dots, \\ \frac{a}{2} + \log \pi - \beta &= 1.6351445123552632926\dots, \\ \kappa_0 &= 3.1514961294723187980\dots, \\ \kappa_1 &= 2.7480700148713340157\dots, \\ \kappa_2 &= 2.6514961294723187980\dots, \\ \kappa_\infty &= 2.3590148791117407073\dots \end{aligned}$$

The LIL conversions are

$$\frac{\pi}{2} \sqrt{2n \log \log n} = \frac{\pi}{\sqrt{2}} \sqrt{n \log \log n},$$

and, because $n \sim (\log x)/\log 2$, fixed-ray LIL fluctuations in the frequency variable have order

$$\sqrt{\log x \log \log \log x}.$$

B Source map

The repository sources audited in this article are listed below so that every claim can be traced to its originating draft. The present article changes two points relative to the comparative audit: it retains the factor- $\sqrt{2}$ LIL correction, and it replaces the unrestricted nonexistence claim by the shell-adaptive analytic existence theorem of Section 7.

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